



Ultra High-Definition Display Controller Family Specification

**Preliminary Version
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1. Features

General

- RTD2796(B) supports input format up to 4096x2160 @ 60Hz.
- RTD2786(B) supports input format up to 2560x1600 @ 60Hz, 2560x2048 @ 50Hz.
- RTD2776(B) supports input format up to 2048x1152 @ 60Hz, 1920x1200 @ 60Hz.
- RTD2796(B) supports one analog RGB input and six multiple-digital-interface combo inputs
- RTD2786(B)/RTD2776(B) supports one analog RGB input and four multiple-digital-interface combo inputs
- Support multiple panel interfaces like LVDS, V-by-One (only RTD2796(B)), and eDP
- Support DisplayPort 1.2 MST daisy-chaining
- RTD27x6 supports PIP / PBP and RTD2796 even supports 4P function
- Zoom scaling up and down
- Embedded one MCU with SPI flash controller.
- It contains 4 ADCs in key pad application
- Require only one crystal to generate all timing.
- Programmable internal low-voltage-reset (LVR)
- High resolution 6 channels PWM output, and wide range selectable PWM frequency.

Crystal

- Support 14.318MHz crystal type

Analog RGB Input Interface

- 1 Analog input supported
- Integrated 8-bit triple-channel 210MHz ADC/PLL
- Embedded programmable Schmitt trigger of HSYNC
- Support Sync-On-Green (SOG) and various kinds of composite sync modes
- On-chip high-performance hybrid PLLs
- High resolution true 64 phase ADC PLL
- YPbPr support up to HDTV 1080p resolution

Ultra-High Speed Combo Reciever

- 2 ports of Ultra-High Speed Combo Reciecers.
- Each port can be configured as HDMI2.0 (6GHz), DisplayPort1.2(5.4GHz, HBR2), MyDP(5.4GHz, HBR2), or DVI as desired
- In HDMI mode, the latest HDMI2.0 is supported
- In HDMI mode, data enable only mode is supported
- In HDMI mode, 6-bit, 8-bit, 10-bit, and 12-bit

color depth transport is supported

- In HDMI mode, High-Bandwidth Digital Content Protection (HDCP 1.4) is supported
- In HDMI mode, HDMI audio is allowed to transmit to I2S/SPDIF output
- In DisplayPort mode, the latest DisplayPort 1.2 is supported
- In DisplayPort mode, three link layer speed HBR2 (5.4GHz), HBR (2.7GHz), RBR (1.62GHz) are supported
- In DisplayPort mode, 6-bit, 8-bit, 10-bit, and 12-bit color depth transport is supported
- In DisplayPort mode, High-Bandwidth Digital Content Protection (HDCP 1.4) is supported
- In DisplayPort mode, HDMI audio is allowed to transmit to I2S/SPDIF output
- In MyDP mode, three link layer speed HBR2 (5.4GHz), HBR (2.7GHz), RBR (1.62GHz) are supported
- In MyDP mode, High-Bandwidth Digital Content Protection (HDCP 1.4) is supported
- In DVI mode, Digital Content Protection (HDCP 1.4) is supported
- In DVI mode, two adjacent receivers to support dual-link DVI with HDCP

High Speed Combo Reciever

- RTD2796(B) supports 4 ports of High Speed Combo Reciecers.
- RTD2786(B)/RTD2776(B) supports 2 ports of High Speed Combo Reciecers.
- Each port can be configured as HDMI1.4 (3GHz), MHL2.1 (3GHz), or DVI as desired
- In HDMI mode, HDMI1.4 is supported
- In HDMI mode, data enable only mode is supported
- In HDMI mode, 6-bit, 8-bit, 10-bit, and 12-bit color depth transport is supported
- In HDMI mode, High-Bandwidth Digital Content Protection (HDCP 1.4) is supported
- In HDMI mode, HDMI audio is allowed to transmit to I2S/SPDIF output
- In MHL mode, MHL2.1 is supported
- In MHL mode, High-Bandwidth Digital Content Protection (HDCP 1.4) is supported
- In MHL mode, packet pixel mode is supported
- In DVI mode, Digital Content Protection (HDCP 1.4) is supported
- In DVI mode, two adjacent receivers to support dual-link DVI with HDCP

Embedded MCU

- Industrial standard 8051 core with external serial flash



- Low speed ADC for various application
- I2C Master or Slave hardware supported

Auto Detection /Auto Calibration

- Input format detection
- Compatibility with standard VESA mode and support user-defined mode
- Smart engine for Phase/Image position/Color calibration

Audio

- Output: IIS , SPDIF
- Embedded Audio DAC
- Embedded headphone amp

Scaling

- Fully programmable zoom ratios
- Independent horizontal/vertical scaling
- Advanced zoom algorithm provides high image quality
- Sharpness/Smooth filter enhancement
- Support non-linear scaling from 4:3 to 16:9 or 16:9 to 4:3

Color Processor

- True 12-bit color processing engine
- Programmable 14-bit gamma support
- Programmable 12-bit 3D gamma support
- xvYCC supported
- Adobe/sRGB compliance
- Advanced dithering logic for the fewer panel color depth enhancement
- Dynamic overshoot-smear canceling engine
- Brightness and contrast control
- Peaking/Coring function for video sharpness
- Support UltraVivid III function to enhance image quality with minimal artificial effect on productivity applications
- Panel Uniformity (Brightness and color uniformity)

VividColor™

- Independent color management (ICM)
- Dynamic contrast control (DCC)
- 2nd generation of Precise color mapping (PCM)
- Content adaptive backlight control (CABC)
- Support ADC Noise Reduction

Embedded DDR3 Controller

- RTD2796(B) support maximal 2 external 16-bit DDR3 DRAM
- RTD2796B/RTD2786(B)/RTD2776(B) supports maximal 1 external 16-bit DDR3 DRAM
- Support DDR3 speed up to 1.6GHz
- Support 90 degree image rotation:

Portrait-to-Landscape or
Landscape-to-Potrait

- LiveShow™ Function, High-performance RTC (response time compensation).
- Frame Rate Control Function
- RTD27x6 supports PIP / PBP and RTD2796 even supports 4P function

Output Interface

- Support 8-bit / 10-bit output through either LVDS, V-by-1(only available for RTD2796(B)), or eDP
- Support 8-port LVDS (available for RTD2796(B)) / 4-port LVDS (available for RTD2786(B)) / 2-port LVDS (available for RTD2776(B)) with the speed of each port up to 100MHz
- RTD2796(B) supports 8-lane V-by-One or 8-lane eDP (HBR) with the output format up to 4k2k (4096x2160 @ 60Hz).
- Support 4-lane eDP (HBR2) with the output format up to 4096x2160 @ 60Hz (available for RTD2796(B)) / 2560x1600 @ 60Hz (available for RTD2786(B)) / 1920x1200 @ 60Hz (available for RTD2776(B)) .
- Support DisplayPort 1.2 Multi-stream Transport (MST) with 3 maximal downstream capability. Share same ports of 4-lane eDP (HBR2)
- Fully programmable display timing generator
- Flexible data pair swapping for easier system design.
- Fixed Last Line output for perfect panel capability

Embedded OSD

- Embedded 128K SRAM dynamically stores OSD command and fonts
- Support multi-color RAM font, 1, 2 and 4-bit per pixel
- 64 color palette
- Maximum 26 window with alpha-blending / gradient / gradient target color / gradient reversed color/ dynamic fade-in/fade-out, bordering/ shadow
- Rotate 90,180,270 degree
- Independent row shadowing/bordering
- Programmable blinking effects for each character
- OSD-made internal pattern generator for factory mode
- Support 12x18 ~ 4x18 proportional font
- Hardware decompression for OSD font
- Support OSD scrolling
- Support 2 independent font based OSD

Power Supply

- 3.3V / 1.5V / 1.1V power supply



2. Ordering Information

Part No.	Max. Resolution/ Timing	Input: VGA	Input: HDMI2/ DP1.2/ MyDP/ DVI Combo Rx	Input: HDMI1.4/ MHL2/ DVI Combo Rx	Output: DP1.2 MST out/ 4-lane HBR2 eDP	Output: LVDS	Output: 8-lane Vx1 / 8-lane HBR eDP	PIP/ PBP	4P	FRC	OD	Max number of DDR3 support	Package
RTD2796-CG	4096x2160@60Hz 3840x2160@60Hz	•	• (2-port)	• (4-port)	•	• (8-port)	•	•	•	•	•	2	702-ball PBGA (green package)
RTD2796B-CG	4096x2160@60Hz 3840x2160@60Hz	•	• (2-port)	• (4-port)	•	• (8-port)	•			•	•	2	702-ball PBGA (green package)
RTD2786-CG	2560x1600@60Hz 2560x2048@50Hz 1920x1080@120Hz	•	• (2-port)	• (2-port)	•	• (4-port)		•		•	•	1	702-ball PBGA (green package)
RTD2786B-CG	2560x1600@60Hz 2560x2048@50Hz 1920x1080@120Hz	•	• (2-port)	• (2-port)	•	• (4-port)				•	•	1	702-ball PBGA (green package)
RTD2776-CG	1920x1200@60Hz	•	• (2-port)	• (2-port)	•	• (2-port)		•		•	•	1	702-ball PBGA (green package)
RTD2776B-CG	1920x1200@60Hz	•	• (2-port)	• (2-port)	•	• (2-port)				•	•	1	702-ball PBGA (green package)



3. Ball Diagram

702-Ball PBGA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32				
A	GD1_GND	GD1_GND	CBUS2	CBUS2	1.5V_DDR	1.5V_DDR		A_VREF	A_CAS2	DGND		DGND	DGND		DGND	A_DQS0		A_DQS0	A_DQ10		A_DQ21	DGND		DGND	A_DQ22		A_DQ18	A_DQ22			TXA1N	TXA0P	TXA0N	A		
B	RX2N_5	RX1P_5	RX2P_5	GD1_GND	1.5V_DDR	1.5V_DDR		A_A13	A_A11	A_A14	DGND	A_CKE	A_MCLKZ	DGND	A_DQ7	DGND	A_DQS1	A_DQS0B	A_DM1	A_DQ6	DGND	A_DQS3B	DGND	A_DQS2	DGND	A_DM3	A_DQ16	A_DQ20			TXA2N	TXA1P		B		
C				GD1_GND	1.5V_DDR	1.5V_DDR		A_A9	DGND	A_A4	A_A10	A_C84	A_MCLK	A_DQS5	A_DM0	A_DQS1B	DGND	DGND	DGND	A_DQ4	A_DQ23	A_DQS3	A_DQS32B	A_DQS2	A_DQS1	A_DQ27	DGND	DGND	DGND			TXACP	TXA2P	TXACN	C	
D	RXC_N_5	RXC_P_5	RX0N_5	GD1_GND	1.5V_DDR	1.5V_DDR	DGN D	A_A5		DGND	A_BA2	A_A3	A_ODT	A_A2	A_A6	A_A8	A_DQ1	A_DQ3	A_DQ12	A_DQ3	A_DQ8	A_DQ24	A_DM2	A_DQ24	A_DQ28	A_DQ25					TXA4N	TXA3P	TXA3N	D		
E	RX2N_4	RX1P_4	RX2P_4	GD1_GND	1.5V_DDR	1.5V_DDR	DGN D	DGND	D_20	D_VREF	DGND	DGND	A_BA8	A_A7	A_RESET	A_RA3Z	A_A1	A_A12	A_BA1	A_DQ10	A_DQ14	A_DQ15	A_DQ13	A_DQ17	A_DQ19	A_DQ26	A_DQ29					TXB0N	TXA4P		E	
F				GD1_GND	1.5V_DDR	1.5V_DDR	DDR_PLL_GND	DDRPLL_VDD01					DGN D	DGN D	DGN D					DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D			TXB1P	TXB0P	TXB1N	F		
G	RXC_N_4	RXC_P_4	RX0N_4	GD1_GND	EEVD								DGN D	DGN D	DGN D					DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D			TXBCN	TXB2P	TXB2N	G		
H	RX2N_3	RX1P_3	RX2P_3	GD1_GND	SCAN_MODE_N	EEI2_CSDA							DGN D								DGN D	DGN D	DGN D	DGN D					TCO_N[0]	TCO_N[1]	TXB3N	TXBCP		H		
J				GD1_GND	DDCSDA5	DDCSCL5	EEI2_CSC1								VCCK_OFF	VCCK_OFF													TCO_N[2]	TCO_N[3]	TXB4P	TXB3P	TXB4N	J		
K	RXC_N_3	RXC_P_3	RX0N_3	GD1_GND	DDCSDA4	DDCSCL4	CEC0								VCCK_OFF		VCCK_OFF												IICS_CL_0	IICS_DA_0	TXC1N	TXC0P	TXC0N	K		
L	RX2N_2	RX1P_2	RX2P_2	GD1_GND	DDCSDA3	DDCSCL3	CEC1								1.5V_DDR	1.5V_DDR	1.5V_DDR	1.5V_DDR	1.5V_DDR	1.5V_DDR									PWM_5_0	PWM_4_0	TXC2N	TXC1P		L		
M				GD1_GND	DDCSDA2	DDCSCL2										1.5V_DDR	1.5V_DDR	1.5V_DDR	1.5V_DDR										PWM_3_0	PWM_2_0	TXCCP	TXC2P	TXCCN	M		
N	RXC_N_2	RXC_P_2	RX0N_2	GD1_GND	MHL_CABLE_DET2	MHL_CABLE_DET3	RESET_B			TMDS_CLK7	GD1_VDD3_3	GD1_VDD3_3		VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D		VCCK_OFF	PVCC	PVCC				PWM_1_0	PWM_0_0	TXC4N	TXC3P	TXC3N	N		
P				LANE0N_1	LANE0P_1	GD1_GND					GD1_GND	GD1_GND		VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	VCCK_OFF	PVCC	PVCC	DGN D	DGN D		SPDIF_F3	SPDIF_F2	TXD0N	TXC4P		P		
R	LANE1N_1	LANE1P_1		GD1_GND	GD1_GND		MHL_VDD33	MHL_VDD33	MHL_VDD33	MHL_VDD33			VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	VCCK_OFF	PVCC	PVCC	DGN D	DGN D	DGN D	SPDIF_F1	SPDIF_F0	TXD1P	TXD0P	TXD1N	R		
T				LANE2N_1	LANE2P_1	GD1_GND					GD1_GND	GD1_GND		VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	VCCK_OFF	PVCC	PVCC	DGN D	DGN D	DGN D	MCK	SCK	TXDCN	TXD2P	TXD2N	T		
U	LANE3N_1	LANE3P_1		GD1_GND	GD1_GND		GD1_VDD10	GD1_VDD10	GD1_VDD10	GD1_VDD10			VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	VCCK_OFF	PVCC	PVCC	DGN D	DGN D	DGN D	WS		TXD3N	TXDCP		U		
V				LANE0N_0	LANE0P_0	GD1_GND					GD1_GND	GD1_GND		VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	VCCK_OFF	VCCK_OFF	VCCK_OFF	DGN D	DGN D	DGN D	DGN D	DGN D		TXD4P	TXD3P	TXD4N	V	
W	LANE1N_0	LANE1P_0		GD1_GND	GD1_GND		GD1_GND	GD1_GND	GD1_GND	GD1_GND			VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	VCCK_OFF	VCCK_OFF	VCCK_OFF	DGN D	DGN D	DGN D	DGN D	DGN D		TXE1N	TXE0P	TXE0N	W	
Y				LANE2N_0	LANE2P_0	GD1_GND	GD1_GND	GD1_GND	GD1_GND	GD1_GND			VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	VCCK_OFF	VCCK_OFF	VCCK_OFF	DGN D	DGN D	DGN D	DGN D	DGN D		TXE2N	TXE1P		Y	
AA	LANE3N_0	LANE3P_0					DDCSDA4	DDCSCL4	DDCSCL4	DDCSCL4			VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	VCCK_OFF	VCCK_OFF	VCCK_OFF	DGN D	DGN D	DGN D	DGN D	DGN D		TXECP	TXE2P	TXECN	AA	
AB				CBUS1	CBUS0	GD1_GND	GD1_GND	GD1_GND	GD1_GND	GD1_GND			VCCK_ON	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	DGN D	VCCK_ON	VCCK_ON	VCCK_ON	VCCK_ON	VCCK_ON		DP_CA_B_DET1_2	PWM2_1	TXE4N	TXE3P	TXE3N	AB			
AC	AVS	AHS	B-	VS_1_OPI0	VS_2_OPI0	VS_0_OPI0					GD1_GND	GD1_GND		AUDIO_VDD33	AUDIO_GND	VTX_VDD10	VTX_GND	VTX_VDD33	VTX_VDD33	VTX_GND	VTX_VDD10		VCCK_ON	VCCK_ON	VCCK_ON	VCCK_ON	VCCK_ON		UART_TX	HDMI_AB_DET2	TXF0N	TXE4P		AC		
AD		B+	G-	ADC_BND	VS_1_OPI0	VS_2_OPI0					ADC_VDD33	ADC_GND		AUDIO_VDD33	AUDIO_GND	VTX_VDD10		VTX_VDD33	VTX_VDD33	VTX_GND	VTX_VDD10		VCCK_ON	VCCK_ON	VCCK_ON	VCCK_ON	VCCK_ON		DP_CA_B_DET1_5	PWM1_1	TXF1P	TXF0P	TXF1N	AD		
AE	SOG	G+	R-	VS_7_OPI0	VS_8_OPI0	VS_6_OPI0					ADC_GND	ADC_GND	ADC_GND			AUDIO_REF													PWM0_1	DP_CA_B_DET0_1	TXFCN	TXF2P	TXF2N	AE		
AF			R+	VOLN_OPI0																									PWM5_1	PWM6_1	TXF3N	TXFCP		AF		
AG	DDCS_DA_VG_A	DDCS_CL_VG_A	IICS_CL_1	IICS_DA_1			HDMI_CAB_DET0						AUDIO_GND	AUDIO_GND	AUDIO_GND														UART_RX	PWM1_1	TXF4P	TXF3P	TXF4N	AG		
AH	MY_DP_SWITC_H_0	MY_DP_SWITC_H_1	HPD_1	HPD_0			DPTX_HPD	VTX_GND	VTX_GND	VTX_GND	VTX_GND	AUDIO_GND	AUDIO_GND		VTX_HPD	VTX_GND	VTX_LL_LOCK	VTX_GND	VTX_GND	VTX_GND	VTX_GND	VTX_GND	VTX_GND	VTX_GND	HDMI_AB_DET1_1	DP_CA_B_DET0_1	DP_CA_B_DET0_2			TXG1N	TXG0P	TXG0N	AH			
AJ	VS_PD	VBA_CAB_DET_0	HPD_2	HPD_2			DPTX_UX_CH_0	DPTX_LAN2P	DPTX_LAN2P	DPTX_LAN2P	DPTX_LAN2P	AUDIO_SOUT_L	AUDIO_SOUT_R		USB_8_PL_CEB1		VTX_T_X7N	VTX_T_X5N	VTX_T_X3N	VTX_T_X1N											TXG2N	TXG1P		AJ		
AK	PANEL_POWER_CTRL	KEY_POWER_CTRL	HPD_5	HPD_4			DPTX_UX_CH_1	DPTX_LAN2N	DPTX_LAN2N	DPTX_LAN2N	DPTX_LAN2N	AUDIO_SOUT_R		FLASH_WP	USB_8_PL_CEB0		VTX_T_X7P	VTX_T_X5P	VTX_T_X3P	VTX_T_X1P										TXH3P	TXHCN	TXH2N	TXGCP	TXG2P	TXGCN	AK
AL	A-ADC1	A-ADC0		GD1_GND			DPTX_LAN3P	DPTX_LAN1P	DPTX_LAN1P	DPTX_LAN1P	LINE_INL	SPL_CLK	SPL_CS	USB_8_PL_CEB1	USB_8_PL_CEB1		VTX_T_X7N	VTX_T_X4N	VTX_T_X2N	VTX_T_X0N							TXH4P	TXH3N	TXH2P	TXH1P	TXH0N	TXG4N	TXG3N	AL		
AM	A-ADC3	A-ADC2	XI	XO			DPTX_LAN3N	DPTX_LAN1N	DPTX_LAN1N	DPTX_LAN1N	LINE_INR	SPL_SI	SPL_SO	USB_8_PL_CEB0	USB_8_PL_CEB0		VTX_T_X7P	VTX_T_X4P	VTX_T_X2P	VTX_T_X0P							TXH4N	TXH3N			TXH1N	TXH0P	TXG4P	TXG3P	AM	

Ball Diagram of RTD2796



4. Table of Pin Assignment

Signals Pin Assignment

(I/O Legend: A = Analog, I = Input, O = Output, P = Power, G = Ground)

Total : 368 balls

BGA Ball Name	I/O	Pin #	RTD2796/RTD2796B Description	RTD2786/RTD2786B Description	RTD2776B Description	Note
CBUS2 / GPIO	AIO	A4	MHL CBUS 2 / MCU GPIO	The same	The same	5V tolerance even when power-off
MHL_CABLE_DET2 / GPIO	IO	N5	MHL Cable Detect 2 / MCU GPIO	The same	The same	5V tolerance even when power-off
CBUS3 / GPIO	AIO	A3	MHL CBUS 3 / MCU GPIO	The same	The same	5V tolerance even when power-off
MHL_CABLE_DET3 / GPIO	IO	N6	MHL Cable Detect 3 / MCU GPIO	The same	The same	5V tolerance even when power-off
TMDS REXT	AI	N9	Impedance Match Reference Resistor For Scan mode, it should be pulled high	The same	The same	Ref value: 12 K ohm (Reference to GND)
RX2P_5	AI	B3	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX2N_5	AI	B1	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX1P_5	AI	B2	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX1N_5	AI	C3	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX0P_5	AI	C2	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX0N_5	AI	D3	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RXCP_5	AI	D2	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RXCN_5	AI	D1	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX2P_4	AI	E3	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX2N_4	AI	E1	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX1P_4	AI	E2	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX1N_4	AI	F3	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX0P_4	AI	F2	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX0N_4	AI	G3	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RXCP_4	AI	G2	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RXCN_4	AI	G1	TMDS Differential Signal Input	NC	NC	3.3V tolerance
RX2P_3	AI	H3	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX2N_3	AI	H1	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX1P_3	AI	H2	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX1N_3	AI	J3	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX0P_3	AI	J2	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX0N_3	AI	K3	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RXCP_3	AI	K2	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RXCN_3	AI	K1	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX2P_2	AI	L3	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX2N_2	AI	L1	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX1P_2	AI	L2	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX1N_2	AI	M3	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX0P_2	AI	M2	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RX0N_2	AI	N3	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RXCP_2	AI	N2	TMDS Differential Signal Input	The same	The same	3.3V tolerance
RXCN_2	AI	N1	TMDS Differential Signal Input	The same	The same	3.3V tolerance
LANE0P_1	AI	P5	DP Input : LANE0P / TMDS Differential Signal Input	The same	The same	3.3V tolerance
LANE0N_1	AI	P4	DP Input : LANE0N / TMDS Differential Signal Input	The same	The same	3.3V tolerance
LANE1P_1	AI	R2	DP Input : LANE1P / TMDS	The same	The same	3.3V tolerance



			Differential Signal Input			
LANE1N_1	AI	R1	DP Input : LANE1N / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE2P_1	AI	T4	DP Input : LANE2P / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE2N_1	AI	T3	DP Input : LANE2N / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE3P_1	AI	U2	DP Input : LANE3P / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE3N_1	AI	U1	DP Input : LANE3N / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE0P_0	AI	V4	DP Input : LANE0P / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE0N_0	AI	V3	DP Input : LANE0N / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE1P_0	AI	W2	DP Input : LANE1P / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE1N_0	AI	W1	DP Input : LANE2N / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE2P_0	AI	Y4	DP Input : LANE2P / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE2N_0	AI	Y3	DP Input : LANE2N / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE3P_0	AI	AA 2	DP Input : LANE3P / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
LANE3N_0	AI	AA 1	DP Input : LANE3N / TMDSDifferential Signal Input	The same	The same	3.3V tolerance
CBUS0 / GPIO	AIO	AB 5	MHL CBUS 0 / MCU GPIO	The same	The same	5V tolerance even when power-off
MHL_CABLE_DET0 / GPIO	I	Y7	MHL Cable Detect 0 / MCU GPIO	The same	The same	5V tolerance even when power-off
CBUS1 / GPIO	AIO	AB 4	MHL CBUS 1/ MCU GPIO	The same	The same	5V tolerance even when power-off
MHL_CABLE_DET1 / GPIO	I	Y6	MHL Cable Detect 1 / MCU GPIO	The same	The same	5V tolerance even when power-off
AUXP_D1 / DDCSCL_D 1	IO	AA 7	AUX-CH 1 / DDC1 (Open drain I/O)	The same	The same	5V tolerance even when power-off
AUXN_D1 / DDCSDA_D 1	IO	AA 6	AUX-CH 1 / DDC1 (Open drain I/O)	The same	The same	5V tolerance even when power-off
AUXP_D0 / DDCSCL_D 0	IO	AB 7	AUX-CH 0 / DDC0 (Open drain I/O)	The same	The same	5V tolerance even when power-off
AUXN_D0 / DDCSDA_D 0	IO	AB 6	AUX-CH 0 / DDC0 (Open drain I/O)	The same	The same	5V tolerance even when power-off
V8_0 / GPIO / Test4b	IO	AC 6	Video 8 input 0 / MCU GPIO	The same	The same	3.3V tolerance
V8_1 / GPIO / Test4b	IO	AC 4	Video 8 input 1 / MCU GPIO	The same	The same	3.3V tolerance
V8_2 / GPIO / Test4b	IO	AC 5	Video 8 input 2 / MCU GPIO	The same	The same	3.3V tolerance
V8_3 / GPIO / Test4b	IO	AD 6	Video 8 input 3 / MCU GPIO	The same	The same	3.3V tolerance
V8_4 / GPIO / Test4b	IO	AD 5	Video 8 input 4 / MCU GPIO	The same	The same	3.3V tolerance
V8_5 / GPIO / Test4b	IO	AE 6	Video 8 input 5 / MCU GPIO	The same	The same	3.3V tolerance
V8_6 / GPIO / Test4b	IO	AE 5	Video 8 input 6 / MCU GPIO	The same	The same	3.3V tolerance



V8_7 / GPIO / Test4b	IO	AE 4	Video 8 input 7 / MCU GPIO	The same	The same	3.3V tolerance
VCLK / GPIO / Test4b	IO	AF 4	Video Clock / MCU GPIO	The same	The same	3.3V tolerance
AVS	I	AC 1	ADC vertical sync input	The same	The same	5V tolerance even when power-off
AHS	I	AC 2	ADC horizontal sync input	The same	The same	5V tolerance even when power-off
B-	AI	AC 3	Negative Blue analog input (Pb-)	The same	The same	3.3V tolerance
B+	AI	AD 2	Positive Blue analog input (Pb+)	The same	The same	3.3V tolerance
G-	AI	AD 3	Negative Green analog input (Y-)	The same	The same	3.3V tolerance
G+	AI	AE 2	Positive Green analog input (Y+)	The same	The same	3.3V tolerance
SOG	AI	AE 1	Sync-On-Green	The same	The same	3.3V tolerance
R-	AI	AE 3	Negative RED analog input (Pr-)	The same	The same	3.3V tolerance
R+	AI	AF 3	Positive RED analog input (Pr+)	The same	The same	3.3V tolerance
DDCSCL_VGA / GPIO	IO	AG 2	DDC(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
DDCSDA_VGA / GPIO	IO	AG 1	DDC1(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
MY_DP_SW ITCH_1 / PWM2 / TCON[7] / GPIO	IO	AH 2	My DP Switch 1 / PWM / TCON / MCU GPIO	The same	The same	5V tolerance even when power-off
MY_DP_SW ITCH_0 / PWM3 / TCON[8] / GPIO	IO	AH 1	My DP Switch 0 / PWM / TCON / MCU GPIO	The same	The same	5V tolerance even when power-off
VGA_CAB_DET_0 / PWM0 / TCON[4] / GPIO	I	AJ2	VGA Cable Detect 0 / PWM / TCON / MCU GPIO	The same	The same	3.3V tolerance
V8_PD / PWM1 / TCON[5] / INT0 / GPIO	I	AJ1	Video8 Power Down / PWM / TCON / INT / MCU GPIO	The same	The same	3.3V tolerance
KEY_POWER_CTL / PWM4 / TCON[10] / INT1 / GPIO	IO	AK 2	Key Power Control / PWM / TCON / MCU GPIO	The same	The same	3.3V tolerance
PANEL_POWER_CTL / PWM5 / TCON[12] / GPIO	IO	AK 1	Panel Power Control / PWM / TCON / MCU GPIO	The same	The same	3.3V tolerance
IICSDA_1 / PWM0 / TCON[3] / GPIO	IO	AG 4	IIC BUS / PWM / TCON / MCU GPIO	The same	The same	5V tolerance even when power-off
IICSCL_1 / PWM1 /	IO	AG 3	IIC BUS / PWM / TCON / MCU GPIO	The same	The same	5V tolerance even when



TTCON[7] / GPIO						power-off
HPD_0 / GPIO	I	AH 4	Hot Plug Detect 0 / MCU GPIO	The same	The same	5V tolerance even when power-off
HPD_1 / GPIO	I	AH 3	Hot Plug Detect 1 / MCU GPIO	The same	The same	5V tolerance even when power-off
HPD_2 / GPIO	I	AJ4	Hot Plug Detect 2 / MCU GPIO	The same	The same	5V tolerance even when power-off
HPD_3 / GPIO	I	AJ3	Hot Plug Detect 3 / MCU GPIO	The same	The same	5V tolerance even when power-off
HPD_4 / GPIO	I	AK 4	Hot Plug Detect 4 / MCU GPIO	The same	The same	5V tolerance even when power-off
HPD_5 / GPIO	I	AK 3	Hot Plug Detect 5 / MCU GPIO	The same	The same	5V tolerance even when power-off
HDMI_CAB_DET0 / GPIO	I	AG 7	HDMI Cable Detect 0 / MCU GPIO	The same	The same	3.3V tolerance
DPTX_HPD / GPIO	I	AH 7	DP Hot Plug Detect / MCU GPIO	The same	The same	3.3V tolerance
A-ADC0 / INT0 / GPIO	IO	AL2	8-bit MCU ADC Input / INT / MCU GPIO	The same	The same	5V tolerance
A-ADC1 / INT1 / GPIO	IO	AL1	8-bit MCU ADC Input / INT / MCU GPIO	The same	The same	5V tolerance
A-ADC2 / GPIO	IO	AM 2	8-bit MCU ADC Input / MCU GPIO	The same	The same	5V tolerance
A-ADC3 / GPIO	IO	AM 1	8-bit MCU ADC Input / MCU GPIO	The same	The same	5V tolerance
DPTX_AUX_CH_P / GPIO	IO	AJ6	DPTX AUX-CH / MCU GPIO	The same	The same	3.3V tolerance
DPTX_AUX_CH_N / GPIO	IO	AK 6	DPTX AUX-CH / MCU GPIO	The same	The same	3.3V tolerance
XI	AI	AM 4	Crystal Input	The same	The same	3.3V tolerance
XO	AO	AM 5	Crystal Output	The same	The same	3.3V tolerance
DPTX_LANE3N_0	AO	AM 7	DP Output : LANE3N	The same	The same	3.3V tolerance
DPTX_LANE3P_0	AO	AL7	DP Output : LANE3P	The same	The same	3.3V tolerance
DPTX_LANE2N_0	AO	AK 8	DP Output : LANE2N	The same	The same	3.3V tolerance
DPTX_LANE2P_0	AO	AJ8	DP Output : LANE2P	The same	The same	3.3V tolerance
DPTX_LANE1N_0	AO	AM 9	DP Output : LANE1N	The same	The same	3.3V tolerance
DPTX_LANE1P_0	AO	AL9	DP Output : LANE1P	The same	The same	3.3V tolerance
DPTX_LANE0N_0	AO	AK 10	DP Output : LANE0N	The same	The same	3.3V tolerance
DPTX_LANE0P_0	AO	AJ1 0	DP Output : LANE0P	The same	The same	3.3V tolerance
LINE_INL / WS / GPIO	AI	AL1 1	LINE-IN / IIS-WS / MCU GPIO	The same	The same	3.3V tolerance
LINE_INR / SCK / GPIO	AI	AM 11	LINE-IN / IIS-SCK / MCU GPIO	The same	The same	3.3V tolerance
AUDIO_REF / MCK /	I	AE 14	Audio Reference Resustance / IIS-MCK / MCU GPIO	The same	The same	3.3V tolerance



GPIO						
AUDIO_SO UTL / SD0 / SPDIF0 / GPIO	AO	AJ1 2	Audio Speaker Output / IIS-SD0 / SPDIF 0 / MCU GPIO	The same	The same	3.3V tolerance
AUDIO_SO UTR / SD1 / SPDIF1 / GPIO	AO	AK 12	Audio Speaker Output / IIS-SD1 / SPDIF 1 / MCU GPIO	The same	The same	3.3V tolerance
AUDIO_HO UTL / SD2 / SPDIF2 / GPIO	AO	AH 13	Audio Headphone Output / IIS-SD2 / SPDIF 2 / MCU GPIO	The same	The same	3.3V tolerance
AUDIO_HO UTR / SD3 / SPDIF3 / GPIO	AO	AJ1 3	Audio Headphone Output / IIS-SD3 / SPDIF 3 / MCU GPIO	The same	The same	3.3V tolerance
SPI_CLK	IO	AL1 3	SPI flash serial clock	The same	The same	3.3V tolerance
SPI_SI	IO	AM 13	SPI flash serial data input	The same	The same	3.3V tolerance
SPI_SO	IO	AM 14	SPI flash serial data output	The same	The same	3.3V tolerance
SPI_CEB	IO	AL1 4	SPI flash chip enable bar	The same	The same	3.3V tolerance
FLASH_WP / GPIO	IO	AK 14	FLASH Write Protect / MCU GPIO	The same	The same	3.3V tolerance
USB_SPI_C LK / CLK0 / GPIO	IO	AL1 5	Serial clock / CLK0 /MCU GPIO	The same	The same	3.3V tolerance
USB_SPI_SI / INT0 / GPIO	IO	AL1 6	Serial data input / INT / MCU GPIO	The same	The same	3.3V tolerance
USB_SPI_S O / INT1 / GPIO	IO	AM 16	Serial data output / INT /MCU GPIO	The same	The same	3.3V tolerance
USB_SPI_C EB0 / IRQB / GPIO	IO	AK 15	SPI chip enable bar 0 /IRQB / MCU GPIO	The same	The same	3.3V tolerance
USB_SPI_C EB1 / T2EX / GPIO	IO	AJ1 5	SPI chip enable bar 1 /T2EX/ MCU GPIO	The same	The same	3.3V tolerance
HDMI_CAB_ DET1 / WS / GPIO / Test4b	IO	AH 24	HDMI Cable Detect 1 / IIS-WS / MCU GPIO	The same	The same	3.3V tolerance
HDMI_CAB_ DET2 / SCK / GPIO / Test4b	IO	AC 29	HDMI Cable Detect 2/ IIS-SCK / MCU GPIO	The same	The same	3.3V tolerance
HDMI_CAB_ DET3 / MCK / GPIO / Test4b	IO	AH 26	HDMI Cable Detect 3/ IIS-MCK / MCU GPIO	The same	The same	3.3V tolerance
DP_CAB_D ET1_0 / TCON[8] / GPIO / Test4b	IO	AD 28	DP Cable Detect 1_0/ TCON / MCU GPIO	The same	The same	3.3V tolerance
DP_CAB_D ET1_1 / SD0 / SPDIF0 / TCON[9] / GPIO	IO	AH 25	DP Cable Detect 1_1 / IIS-SD0 / SPDIF0 / TCON / MCU GPIO	The same	The same	3.3V tolerance
DP_CAB_D ET1_2 / SD1	IO	AB 28	DP Cable Detect 1_2 / IIS-SD1 / SPDIF1 / TCON / MCU GPIO	The same	The same	3.3V tolerance



/ SPDIF1 / TCON[10] / GPIO						
DP_CAB_D ET0_0 / SD2 / SPDIF2 / TCON[11] / GPIO	IO	AH 27	DP_CAB_DET0_0 / SD2 / SPDIF2 / TCON / MCU GPIO	The same	The same	3.3V tolerance
DP_CAB_D ET0_1 / SD3 / SPDIF3 / TCON[12] / GPIO	IO	AE 29	DP_CAB_DET0_1 / SD3 / SPDIF3 / TCON / MCU GPIO	The same	The same	3.3V tolerance
DP_CAB_D ET0_2 / TCON[13] / GPIO / Test4b	IO	AH 28	DP_CAB_DET0_2 / TCON / MCU GPIO / Test4b	The same	The same	3.3V tolerance
UART_TX / TCON[0] / GPIO / Test4b	IO	AC 28	UART TX / TCON / MCU GPIO	The same	The same	5V tolerance even when power-off
UART_RX / TCON[1] / GPIO / Test4b	IO	AG 28	UART RX / TCON / MCU GPIO	The same	The same	5V tolerance even when power-off
PWM0_1 / TCON[2] / GPIO / PCB_POWE _DOWN / Test4b	IO	AE 28	PWM / TCON / MCU GPIO / PCB Power Down	The same	The same	5V tolerance even when power-off
PWM1_1 / TCON[3] / GPIO / Test4b	IO	AG 29	PWM / TCON / MCU GPIO	The same	The same	5V tolerance even when power-off
PWM2_1 / TCON[4] / IR_RECEIV ER / GPIO	IO	AB 29	PWM / TCON / IR Receiver / MCU GPIO	The same	The same	5V tolerance even when power-off
PWM3_1 / TCON[5] / T2 / GPIO	IO	AF 29	PWM / TCON / T2 / MCU GPIO	The same	The same	5V tolerance even when power-off
PWM4_1 / TCON[6] / T1 / GPIO	IO	AD 29	PWM / TCON / T1 / MCU GPIO	The same	The same	5V tolerance even when power-off
PWM5_1 / TCON[7] / T0 / GPIO	IO	AF 28	PWM / TCON / T0 / MCU GPIO	The same	The same	5V tolerance even when power-off
VTX_HPD / DPTX_HPD _1 / GPIO	IO	AH 16	V-by-One Hot Plug Detect / DPTX Hot Plug Detect 1 / MCU GPIO	The same	The same	5V tolerance even when power-off
VTX_PLL_L OCK / DPTX_HPD _2 / GPIO	IO	AH 18	V-by-One PLL Lock / DPTX Hot Plug Detect 2 / MCU GPIO	The same	The same	5V tolerance even when power-off
VTX_TX7N / DPTX_LAN E3N_1	AO	AJ1 7	V-by-One Output : 7N / DP Output :LANE3N	NC	NC	3.3V tolerance
VTX_TX7P / DPTX_LAN E3P_1	AO	AK 17	V-by-One Output : 7P / DP Output :LANE3P	NC	NC	3.3V tolerance
VTX_TX6N / DPTX_LAN E2N_1	AO	AL1 8	V-by-One Output : 6N / DP Output :LANE2N	NC	NC	3.3V tolerance
VTX_TX6P /	AO	AM	V-by-One Output : 6P / DP	NC	NC	3.3V tolerance



DPTX_LAN E2P_1		18	Output :LANE2P			
VTX_TX5N / DPTX_LAN E1N_1	AO	AJ1 9	V-by-One Output : 5N / DP Output :LANE1N	NC	NC	3.3V tolerance
VTX_TX5P / DPTX_LAN E1P_1	AO	AK 19	V-by-One Output : 5P / DP Output :LANE1P	NC	NC	3.3V tolerance
VTX_TX4N / DPTX_LAN E0N_1	AO	AL2 0	V-by-One Output : 4N / DP Output :LANE0N	NC	NC	3.3V tolerance
VTX_TX4P / DPTX_LAN E0P_1	AO	AM 20	V-by-One Output : 4P / DP Output :LANE0P	NC	NC	3.3V tolerance
VTX_TX3N / DPTX_LAN E3N_2	AO	AJ2 1	V-by-One Output : 3N / DP Output :LANE3N	NC	NC	3.3V tolerance
VTX_TX3P / DPTX_LAN E3P_2	AO	AK 21	V-by-One Output : 3P / DP Output :LANE3P	NC	NC	3.3V tolerance
VTX_TX2N / DPTX_LAN E2N_2	AO	AL2 2	V-by-One Output : 2N / DP Output :LANE2N	NC	NC	3.3V tolerance
VTX_TX2P / DPTX_LAN E2P_2	AO	AM 22	V-by-One Output : 2P / DP Output :LANE2P	NC	NC	3.3V tolerance
VTX_TX1N / DPTX_LAN E1N_2	AO	AJ2 3	V-by-One Output : 1N / DP Output :LANE1N	NC	NC	3.3V tolerance
VTX_TX1P / DPTX_LAN E1P_2	AO	AK 23	V-by-One Output : 1P / DP Output :LANE1P	NC	NC	3.3V tolerance
VTX_TX0N / DPTX_LAN E0N_2	AO	AL2 4	V-by-One Output : 0N / DP Output :LANE0N	NC	NC	3.3V tolerance
VTX_TX0P / DPTX_LAN E0P_2	AO	AM 24	V-by-One Output : 0P / DP Output :LANE0P	NC	NC	3.3V tolerance
TXH4P	IO	AL2 6	LVDS 10bit Output	NC	NC	3.3V tolerance
TXH4N	IO	AM 26	LVDS 10bit Output	NC	NC	3.3V tolerance
TXH3P / TXH3P_8b	IO	AK 27	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXH3N / TXH3N_8b	IO	AM 27	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXHCP / TXHCP_8b	IO	AL2 7	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXHCN / TXHCN_8b	IO	AK 28	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXH2P / TXH2P_8b	IO	AL2 8	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXH2N / TXH2N_8b	IO	AK 29	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXH1P / TXH1P_8b	IO	AL2 9	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXH1N / TXH1N_8b	IO	AM 29	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXH0P / TXH0P_8b	IO	AM 30	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXH0N / TXH0N_8b	IO	AL3 0	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXG4P	IO	AM 31	LVDS 10bit Output	NC	NC	3.3V tolerance
TXG4N	IO	AL3	LVDS 10bit Output	NC	NC	3.3V tolerance



		1				
TXG3P / TXG3P_8b	IO	AM 32	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXG3N / TXG3N_8b	IO	AL3 2	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXGCP / TXGCP_8b	IO	AK 30	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXGCN / TXGCN_8b	IO	AK 32	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXG2P / TXG2P_8b	IO	AK 31	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXG2N / TXG2N_8b	IO	AJ3 0	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXG1P / TXG1P_8b	IO	AJ3 1	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXG1N / TXG1N_8b	IO	AH 30	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXG0P / TXG0P_8b	IO	AH 31	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXG0N / TXG0N_8b	IO	AH 32	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXF4P	IO	AG 30	LVDS 10bit Output	The same	The same	3.3V tolerance
TXF4N	IO	AG 32	LVDS 10bit Output	The same	The same	3.3V tolerance
TXF3P / TXF3P_8b	IO	AG 31	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXF3N / TXF3N_8b	IO	AF 30	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXFCP / TXFCP_8b	IO	AF 31	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXFCN / TXFCN_8b	IO	AE 30	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXF2P / TXF2P_8b	IO	AE 31	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXF2N / TXF2N_8b	IO	AE 32	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXF1P / TXF1P_8b	IO	AD 30	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXF1N / TXF1N_8b	IO	AD 32	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXF0P / TXF0P_8b	IO	AD 31	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXF0N / TXF0N_8b	IO	AC 30	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXE4P	IO	AC 31	LVDS 10bit Output	The same	The same	3.3V tolerance
TXE4N	IO	AB 30	LVDS 10bit Output	The same	The same	3.3V tolerance
TXE3P / TXE3P_8b	IO	AB 31	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXE3N / TXE3N_8b	IO	AB 32	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXECP / TXECP_8b	IO	AA 30	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXECN / TXECN_8b	IO	AA 32	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXE2P / TXE2P_8b	IO	AA 31	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXE2N / TXE2N_8b	IO	Y30	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXE1P / TXE1P_8b	IO	Y31	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXE1N / TXE1N_8b	IO	W3 0	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance



TXE0P / TXE0P_8b	IO	W3 1	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXE0N / TXE0N_8b	IO	W3 2	LVDS 10bit Output / LVDS 8bit Output	The same	The same	3.3V tolerance
TXD4P	IO	V30	LVDS 10bit Output	The same	NC	3.3V tolerance
TXD4N	IO	V32	LVDS 10bit Output	The same	NC	3.3V tolerance
TXD3P / TXD3P_8b	IO	V31	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXD3N / TXD3N_8b	IO	U3 0	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXDCP / TXDCP_8b	IO	U3 1	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXDCN / TXDCN_8b	IO	T30	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXD2P / TXD2P_8b	IO	T31	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXD2N / TXD2N_8b	IO	T32	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXD1P / TXD1P_8b	IO	R3 0	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXD1N / TXD1N_8b	IO	R3 2	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXD0P / TXD0P_8b	IO	R3 1	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXD0N / TXD0N_8b	IO	P30	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXC4P	IO	P31	LVDS 10bit Output	The same	NC	3.3V tolerance
TXC4N	IO	N3 0	LVDS 10bit Output	The same	NC	3.3V tolerance
TXC3P / TXC3P_8b	IO	N3 1	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXC3N / TXC3N_8b	IO	N3 2	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXCCP / TXCCP_8b	IO	M3 0	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXCCN / TXCCN_8b	IO	M3 2	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXC2P / TXC2P_8b	IO	M3 1	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXC2N / TXC2N_8b	IO	L30	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXC1P / TXC1P_8b	IO	L31	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXC1N / TXC1N_8b	IO	K30	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXC0P / TXC0P_8b	IO	K31	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXC0N / TXC0N_8b	IO	K32	LVDS 10bit Output / LVDS 8bit Output	The same	NC	3.3V tolerance
TXB4P	IO	J30	LVDS 10bit Output	NC	NC	3.3V tolerance
TXB4N	IO	J32	LVDS 10bit Output	NC	NC	3.3V tolerance
TXB3P / TXB3P_8b	IO	J31	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXB3N / TXB3N_8b	IO	H3 0	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXBCP / TXBCP_8b	IO	H3 1	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXBCN / TXBCN_8b	IO	G3 0	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXB2P / TXB2P_8b	IO	G3 1	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXB2N / TXB2N_8b	IO	G3 2	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXB1P / TXB1P_8b	IO	F30	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance



TXB1N / TXB1N_8b	IO	F32	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXB0P / TXB0P_8b	IO	F31	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXB0N / TXB0N_8b	IO	E30	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXA4P	IO	E31	LVDS 10bit Output	NC	NC	3.3V tolerance
TXA4N	IO	D3 0	LVDS 10bit Output	NC	NC	3.3V tolerance
TXA3P / TXA3P_8b	IO	D3 1	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXA3N / TXA3N_8b	IO	D3 2	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXACP / TXACP_8b	IO	C3 0	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXACN / TXACN_8b	IO	C3 2	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXA2P / TXA2P_8b	IO	C3 1	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXA2N / TXA2N_8b	IO	B30	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXA1P / TXA1P_8b	IO	B31	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXA1N / TXA1N_8b	IO	A30	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXA0P / TXA0P_8b	IO	A31	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
TXA0N / TXA0N_8b	IO	A32	LVDS 10bit Output / LVDS 8bit Output	NC	NC	3.3V tolerance
WS / TCON[12] / GPIO / Test4b	IO	U2 8	IIS-WS / TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
SCK / TCON[13] / GPIO / Test4b	IO	T29	IIS-SCK / TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
MCK / GPIO / Test4b	IO	T28	IIS-MCK / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
SPDIF0 / SD0 / GPIO / Test4b	IO	R2 9	SPDIF0 / IIS-SD0 / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
SPDIF1 / SD1 / GPIO / Test4b	IO	R2 8	SPDIF1 / IIS-SD1 / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
SPDIF2 / SD2 / GPIO / Test4b	IO	P29	SPDIF2 / IIS-SD2 / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
SPDIF3 / SD2 / GPIO / Test4b	IO	P28	SPDIF3 / IIS-SD2 / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
DPTX_AUX_CH_P_1 / PWM0_0 / TCON[6] / GPIO	IO	N2 9	DPTX AUX-CH / PWM / TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	5V tolerance even when power-off
DPTX_AUX_CH_N_1 / PWM1_0 / TCON[7] / GPIO	IO	N2 8	DPTX AUX-CH / PWM / TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	5V tolerance even when power-off
DPTX_AUX_CH_P_2 / PWM2_0 /	IO	M2 9	DPTX AUX-CH / PWM / TCON / MCU GPIO (This pin can not work when	The same	The same	5V tolerance even when power-off



TCON[8] / GPIO			power saving & power Down.)			
DPTX_AUX_CH_N_2 / PWM3_0 / TCON[9] / GPIO	IO	M28	DPTX AUX-CH / PWM / TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	5V tolerance even when power-off
DVS / PWM4_0 / TCON[10] / GPIO	IO	L29	DVS / PWM / TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	5V tolerance even when power-off
XTAL_CLK_OUT / PWM5_0 / TCON[11] / GPIO	IO	L28	XTAL_CLK_OUT / PWM / TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	5V tolerance even when power-off
IICSDA_0 / TCON[5] / GPIO / Test4b	IO	K29	IIC BUS / TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	5V tolerance even when power-off
IICSCL_0 / TCON[4] / GPIO / Test4b	IO	K28	IIC BUS / TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	5V tolerance even when power-off
TCON[3] / GPIO / Test4b	IO	J29	TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
TCON[2] / GPIO / Test4b	IO	J28	TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
TCON[1] / GPIO / Test4b	IO	H29	TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
TCON[0] / GPIO / Test4b	IO	H28	TCON / MCU GPIO (This pin can not work when power saving & power Down.)	The same	The same	3.3V tolerance
D_VREF	I	E10	Reference Voltage	The same	The same	
A_DQ0	IO	D22	Data Input / Output	The same	The same	
A_DQ1	IO	D18	Data Input / Output	The same	The same	
A_DQ2	IO	D23	Data Input / Output	The same	The same	
A_DQ3	IO	D19	Data Input / Output	The same	The same	
A_DQ4	IO	C20	Data Input / Output	The same	The same	
A_DQ5	IO	C14	Data Input / Output	The same	The same	
A_DQ6	IO	B20	Data Input / Output	The same	The same	
A_DQ7	IO	B15	Data Input / Output	The same	The same	
A_DQ8	IO	A16	Data Input / Output	The same	The same	
A_DQ9	IO	D21	Data Input / Output	The same	The same	
A_DQ10	IO	E20	Data Input / Output	The same	The same	
A_DQ11	IO	A19	Data Input / Output	The same	The same	
A_DQ12	IO	D20	Data Input / Output	The same	The same	
A_DQ13	IO	E23	Data Input / Output	The same	The same	
A_DQ14	IO	E21	Data Input / Output	The same	The same	
A_DQ15	IO	E22	Data Input / Output	The same	The same	
A_DQ16	IO	B27	Data Input / Output	NC	NC	
A_DQ17	IO	E24	Data Input / Output	NC	NC	
A_DQ18	IO	A27	Data Input / Output	NC	NC	
A_DQ19	IO	E25	Data Input / Output	NC	NC	



A_DQ20	IO	B28	Data Input / Output	NC	NC	
A_DQ21	IO	A21	Data Input / Output	NC	NC	
A_DQ22	IO	A28	Data Input / Output	NC	NC	
A_DQ23	IO	C2 1	Data Input / Output	NC	NC	
A_DQ24	IO	D2 5	Data Input / Output	NC	NC	
A_DQ25	IO	D2 7	Data Input / Output	NC	NC	
A_DQ26	IO	E26	Data Input / Output	NC	NC	
A_DQ27	IO	C2 5	Data Input / Output	NC	NC	
A_DQ28	IO	D2 6	Data Input / Output	NC	NC	
A_DQ29	IO	A25	Data Input / Output	NC	NC	
A_DQ30	IO	E27	Data Input / Output	NC	NC	
A_DQ31	IO	C2 4	Data Input / Output	NC	NC	
A_DM0	IO	C1 5	Input Data Mask	The same	The same	
A_DM1	IO	B19	Input Data Mask	The same	The same	
A_DM2	IO	D2 4	Input Data Mask	The same	The same	
A_DM3	IO	B26	Input Data Mask	The same	The same	
A_DQS0	IO	A18	Data strobe : Output with read data. Edge-aligned with read data. Input with write data. Center-aligned to write data.	The same	The same	
A_DQS0B	IO	B18	Data strobe : Output with read data. Edge-aligned with read data. Input with write data. Center-aligned to write data.	The same	The same	
A_DQS1	IO	B17	Data strobe : Output with read data. Edge-aligned with read data. Input with write data. Center-aligned to write data.	The same	The same	
A_DQS1B	IO	C1 6	Data strobe : Output with read data. Edge-aligned with read data. Input with write data. Center-aligned to write data.	The same	The same	
A_DQS2	IO	B24	Data strobe : Output with read data. Edge-aligned with read data. Input with write data. Center-aligned to write data.	The same	The same	
A_DQS2B	IO	C2 3	Data strobe : Output with read data. Edge-aligned with read data. Input with write data. Center-aligned to write data.	The same	The same	
A_DQS3	IO	C2 2	Data strobe : Output with read data. Edge-aligned with read data. Input with write data. Center-aligned to write data.	The same	The same	
A_DQS3B	IO	B22	Data strobe: Output with read data. Edge-aligned with read data. Input with write data. Center-aligned to write data.	The same	The same	
A_BA0	IO	E13	Bank Address Input	The same	The same	
A_BA1	IO	E19	Bank Address Input	The same	The same	
A_BA2	IO	D11	Bank Address Input	The same	The same	
A_A0	IO	D1 5	Address Input	The same	The same	
A_A1	IO	E17	Address Input	The same	The same	
A_A2	IO	D1 4	Address Input	The same	The same	
A_A3	IO	D1	Address input	The same	The same	



		2				
A_A4	IO	C10	Address Input	The same	The same	
A_A5	IO	D8	Address Input	The same	The same	
A_A6	IO	D16	Address Input	The same	The same	
A_A7	IO	D17	Address Input	The same	The same	
A_A8	IO	D17	Address Input	The same	The same	
A_A9	IO	C8	Address Input	The same	The same	
A_A10	IO	C11	Address Input	The same	The same	
A_A11	IO	B9	Address Input	The same	The same	
A_A12	IO	E18	Address Input	The same	The same	
A_A13	IO	B8	Address Input	The same	The same	
A_A14	IO	B10	Address Input	The same	The same	
A_RESET	IO	E15	Reset : A_RESET is an active LOW CMOS input referenced to VSS.	The same	The same	
A_MCLK	IO	C13	Clock : A_MCLK and A_MCLKZ are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of A_MCLK and negative edge of A_MCLKZ.	The same	The same	
A_MCLKZ	IO	B13	Clock : A_MCLK and A_MCLKZ are differential clock inputs. All address and control input signals are sampled on the crossing of the positive edge of A_MCLK and negative edge of A_MCLKZ.	The same	The same	
A_ODT	IO	D13	On-die termination : ODT enables (registered HIGH) and disables (registered LOW) termination resistance internal to the DDR3 SDRAM.	The same	The same	
A_CKE	IO	B12	Clock enable	The same	The same	
A_CS#	IO	C12	Chip select	The same	The same	
A_RASZ	IO	E16	Command inputs : A_RASZ, A_CASZ, and A_WEZ (along with A_CS#) define the command being entered and are referenced to VREFCA.	The same	The same	
A_CASZ	IO	A9	Command inputs : A_RASZ, A_CASZ, and A_WEZ (along with A_CS#) define the command being entered and are referenced to VREFCA.	The same	The same	
A_WEZ	IO	A8	Command inputs : A_RASZ, A_CASZ, and A_WEZ (along with A_CS#) define the command being entered and are referenced to VREFCA.	The same	The same	
D_ZQ	I	E9	External reference ball for output drive calibration: This ball is tied to an external 240Ω resistor (1%), which is tied to VSSQ.	The same	The same	
CEC1 / GPIO	IO	L7	CEC 1 / MCU GPIO	The same	The same	5V tolerance even when



						power-off
CEC0 / GPIO	IO	K7	CEC 0 / MCU GPIO	The same	The same	5V tolerance even when power-off
EEWD / PWM2 / Tcon[3] / GPIO	IO	G5	EEWD / PWM / Tcon / MCU GPIO	The same	The same	3.3V tolerance
EEI2CSCL / PWM0 / TCON[1] / GPIO	IO	J7	CSCL / PWM / TCON / MCU GPIO	The same	The same	3.3V tolerance
EEI2CSDA / PWM1 / TCON[2] / GPIO	IO	H6	EEI2CSDA / PWM / TCON / MCU GPIO	The same	The same	3.3V tolerance
SCAN_MOD E_N	IO	H5	When AC power is turned on, this ball must be pull "High".	The same	The same	3.3V tolerance
DDCSCL5 / GPIO	IO	J6	DDC5(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
DDCSDA5 / GPIO	IO	J5	DDC5(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
DDCSCL4 / GPIO	IO	K6	DDC4(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
DDCSDA4 / GPIO	IO	K5	DDC4(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
DDCSCL3 / GPIO	IO	K6	DDC4(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
DDCSDA3 / GPIO	IO	K5	DDC4(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
DDCSCL2 / GPIO	IO	K6	DDC4(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
DDCSDA2 / GPIO	IO	K5	DDC4(Open drain I/O) / MCU GPIO	The same	The same	5V tolerance even when power-off
RESETB	I	N7	Chip reset bar	The same	The same	Low active; 5V tolerance even when power-off



Power / Ground Pin Assignment

(I/O Legend: A = Analog, I = Input, O = Output, P = Power, G = Ground)

Total : 334 balls

BGA Ball Name	I/O	Pin #	Description	Note
GDI_GND	AG	A1,A2,B4,C4,D4,E4,F4,G4,H4,J4,K4,L4,M4,N4,P6,P10,P11,R5,R6,T5,T6,T8,T9,T10,T11,U5,U6,V5,V6,W5,W6,W8,W9,W10,W11,Y5,Y10,Y11,AC10,AC11,AL4	Analog Ground	41 balls
GDI_VDD33	AP	N10,N11,AA11,AB10,AB11	3.3V Analog Power	5 balls
GDI_VDD10	AP	U8,U9,U10,U11,V8,V9,V10,V11	1.1V Analog Power	8 balls
MHL_VDD33	AP	R8,R9,R10,R11,AA9,AA10	3.3V Analog Power	6 balls
ADC_VDD33	AP	AD10,AD11	3.3V ADC Power	2 balls
ADC_GND	AG	AD4,AE8,AE9,AE10,AE11	ADC Gound	5 balls
AUDIO_VDD33	AP	AC13,AD13	3.3V Audio Power	2 balls
AUDIO GND	AG	AC14,AD14,AG12,AG13,AG14,AH12,AH14	Audio Gound	7 balls
VTX_VDD10	AP	AB20,AC15,AC20,AD15,AD20	1.1V Analog Power	5 balls
VTX_VDD33	AP	AC17,AC18,AD17,AD18	3.3V Analog Power	4 balls
VTX_GND	AG	AB21,AC16,AC19,AD21,AH8,AH9,AH10,AH11,AH17,AH19,AH20,AH21,AH22,AH23	Analog Ground	14 balls
PVCC	P	N23,N24,P23,P24,R23,R24,T23,T24,U23,U24	Pad Power	10 balls
1.5V_DDR	P	A5,A6,B5,B6,C5,C6,D5,D6,E5,E6,F5,F6,L15,L16,L17,L18,L19,L20,M16,M17,M18,M19	1.5V DDR3 Power	22 balls
VCCK_ON	P	N13,P13,R13,T13,U13,V13,W13,Y13,AA13,AB22,AB23,AB24,AB25,AB26,AC22,AC23,AC24,AC25,AC26,AD22,AD23,AD24,AD25,AD26	1.1V Core Power	24 balls
VCCK_OFF	P	J15,J16,K16,K18,N22,P22,R22,T22,U22,V22,V23,V24,W22,W23,W24,Y22,Y23,Y24,AA22,AA23,AA24	1.1V Core Power	21balls
DDRPLL_VDD10	P	F8	1.1V DDR PLL Power	1 balls
DDR PLL GND	G	F7	DDR PLL Ground	1 balls
DGND	G	A10,A12,A13,A15,A22,A24,B11,B14,B16,B21,B23,B25,C9,C17,C18,C19,C26,C27,C28,D7,D10,E7,E8,E11,E12,F13,F14,F15,F20,F21,F22,F23,F24,F25,F26,F27,F28,G13,G14,G15,G20,G21,G22,G23,G24,G25,G26,G27,G28,H13,H20,H21,H22,H23,N14,N15,N16,N17,N18,N19,N20,P14,P15,P16,P17,P18,P19,P20,P21,P25,P26,R14,R15,R16,R17,R18,R19,R20,R21,R25,R26,R27,T14,T15,T16,T17,T18,T	Digital Ground	156 balls



		19, T20, T21, T25, T26, T27, U14, U15, U16, U17, U18, U19, U20, U21, U25, U26, U27, V14, V15, V16, V17, V18, V19, V20, V21, V25, V26, V27, V28, V29, W14, W15, W16, W17, W18, W19, W20, W21, W25, W26, W27, W28, W29, Y14, Y15, Y16, Y17, Y18, Y19, Y20, Y21, Y25, Y26, Y27, Y28, Y29, A14, AA15, AA16, AA17, AA18, AA19, AA20, AA21, AA25, AA26, AA27, AA28, AA29,		
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5. Electric Specification

DC Characteristics

Recommended Operating Conditions

PARAMETER	SYMBOL	MIN	TYP	MAX	UNITS
Voltage on Input (5V tolerance)	V _{IN}	-1		5	V
Supply Voltage	PVCC	3.14	3.30	3.47	V
Electrostatic Discharge	V _{ESD}			±2.5	kV
Latch-Up	I _{LA}			±100	mA
Ambient Operating Temperature	T _A	0		70	°C
Storage Temperature (plastic)	T _{STG}	-55		110	°C
Thermal Resistance (Junction to Air)	θ _{JA}		14.6		°C/W
Thermal Resistance (Junction to Case)	θ _{JC}		3.9		°C/W
Junction Acceptable Temperature	T _j			110	°C

Absolute Maximum Ratings

PARAMETER	SYMBOL	MIN	TYP	MAX	UNITS
Supply Voltage	PVCC			3.6	V
Storage Temperature (plastic)	T _{STG}			125	°C
Junction Acceptable Temperature	T _j			125	°C

Note : Operation under the absolute maximum ratings does not imply well-functioning. Long-term stress to the absolute maximum ratings would probably affect the device reliability or further cause permanent damage.

PARAMETER	SYMBOL	MIN	TYP	MAX	UNITS
Reset Pulse Period	Trst-en ¹	1120			ns
Power-on-Reset Period	Tpor-rst ²	145	146.5	148	ms

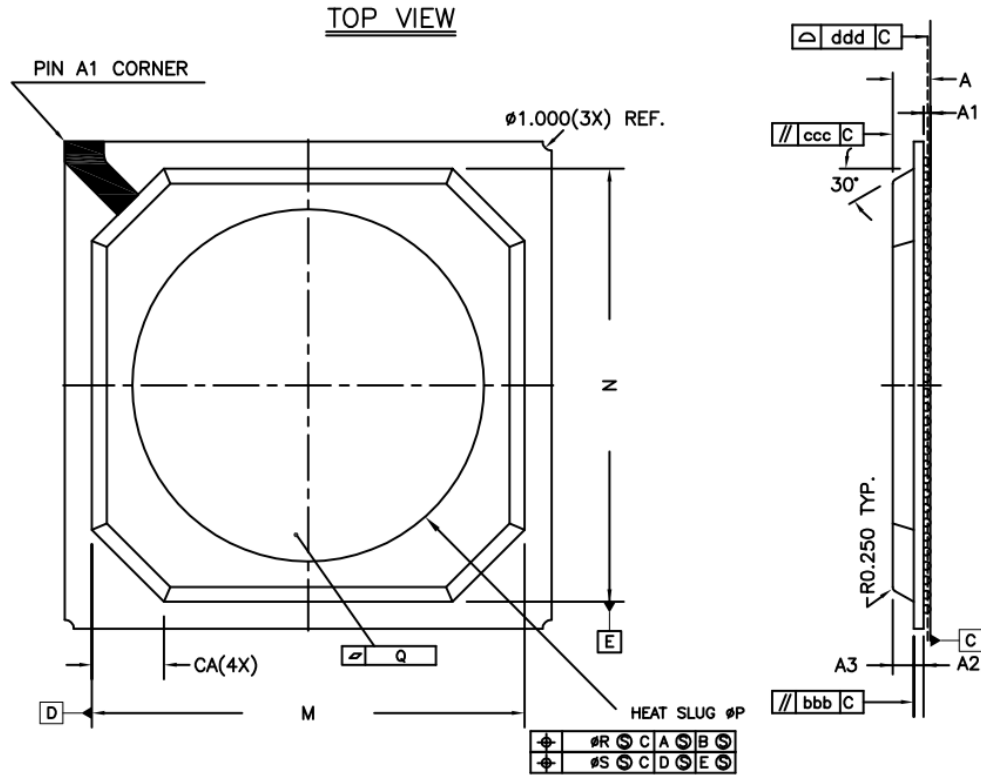
1. $16 * \text{Xtal_cycle}(1/14.3\text{Mhz})$

2. $65536 * 16 * 2 * \text{Xtal_cycle}(1/14.3\text{Mhz})$

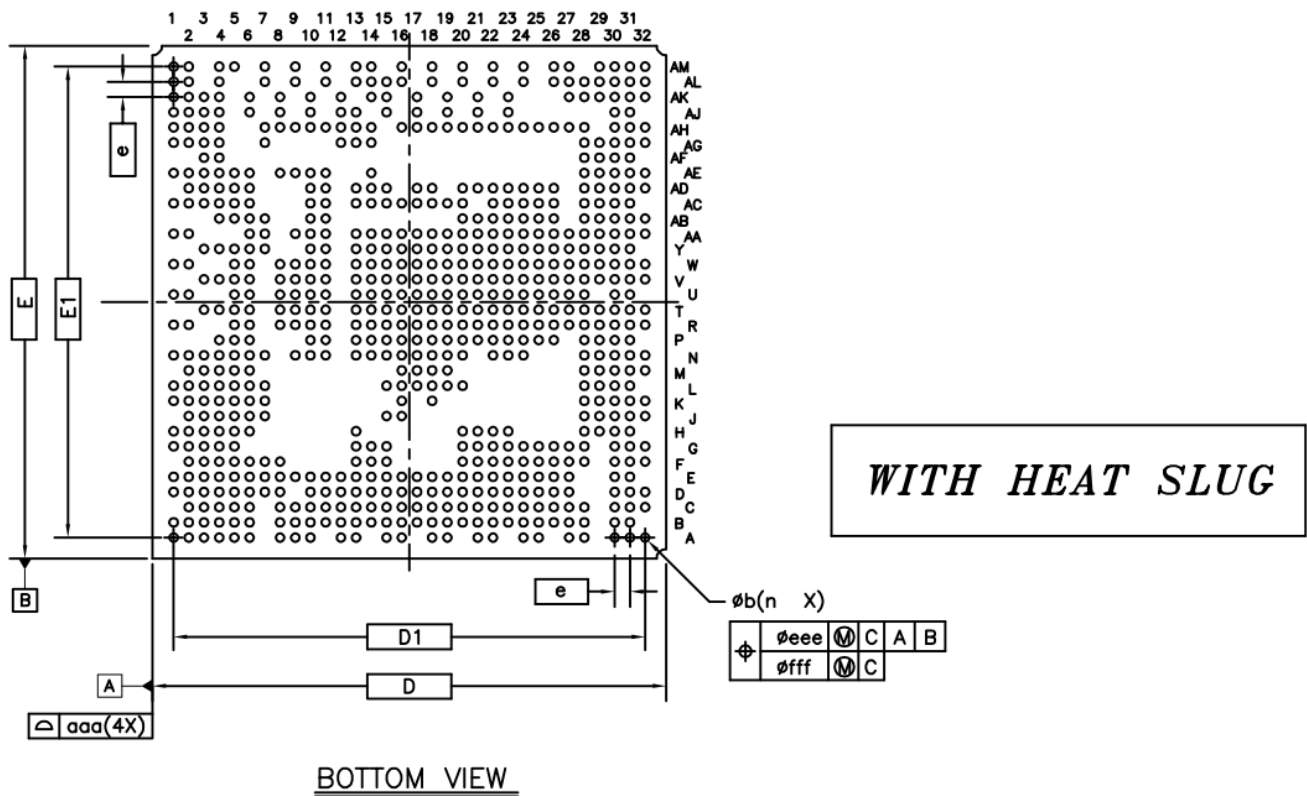


6. Mechanical Specification

Top view



Bottom view





Mechanical dimensions notes

		Symbol	Common Dimensions		
			MIN.	NOM.	MAX.
Package :			HS FBGA		
Body Size:	X	D	27.000		
	Y	E	27.000		
Ball Pitch :		e	0.800		
Total Thickness :		A	1.982	2.100	2.218
Mold Thickness :		A3	1.170	Ref.	
Substrate Thickness :		A2	0.560	Ref.	
Ball Diameter :			0.450		
Stand Off :		A1	0.320	-	0.420
Ball Width :		b	0.375	-	0.525
Mold Area :	X	M	24.000		
	Y	N	24.000		
H/S Exposed Size:		P	19 ~ 20		
H/S Flatness		Q	0.100		
H/S Shift With Substrate Edge:		R	0.300		
H/S Shift With Mold Area:		S	0.500		
Chamfer		CA	4.000	Ref.	
Package Edge Tolerance :		aaa	0.150		
Substrate Parallelism :		bbb	0.100		
Mold Parallelism :		ccc	0.200		
Coplanarity:		ddd	0.150		
Ball Offset (Package) :		eee	0.150		
Ball Offset (Ball) :		fff	0.080		
Ball Count :		n	702		
Edge Ball Center to Center :	X	D1	24.800		
	Y	E1	24.800		

Unit in millimeter “mm”